9 **(a)** Fig. 9.1 shows a transformer.

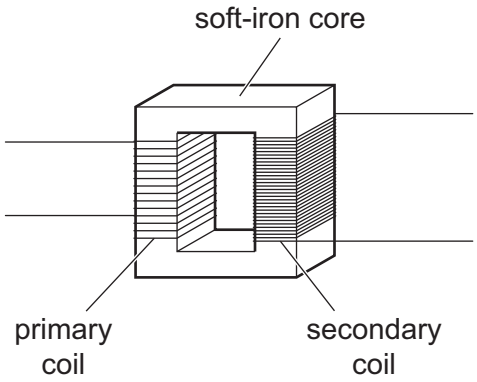

Fig. 9.1

(i) There is an alternating current in the primary coil.

Describe how an alternating current is produced in the secondary coil.

.....

.....

.....

..... [3]

(ii) A step-up transformer has a turns ratio of 1 : 20. The voltage across the primary coil is 12 V.

Calculate the voltage across the secondary coil.

voltage across secondary coil = ... [2]

(b) The power lost in a cable is 1.25×10^{-3} W. The resistance of the cable is 0.050 Ω.

Calculate the current in the cable.

current = ... [2]

(c) State **two** advantages of high-voltage transmission.

1 .....

.....

2 .....

.....

[2]

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[Total: 9]

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